

WILL Project - Evaluation Contribution

1. Group ID

Group 80

2. GitHub Branch

animish-evaluation

3. Evaluation Role

Research/data testing and evaluation support for the WILL-RAG pipeline.

4. Evaluation Aim

Test the RAG pipeline with an additional research paper, verify retrieval quality and source metadata, document errors, and prepare the remaining answer-generation evaluation once API credits are available.

5. Test Dataset

Field	Details
Paper ID	P05
Title	The Attention System of the Human Brain: 20 Years After
Authors	Steven E. Petersen; Michael I. Posner
Year / Journal	2012 / Annual Review of Neuroscience
DOI	10.1146/annurev-neuro-062111-150525
Topic	attention

The raw PDF was kept local and was not intended to be pushed to the public repository.

6. Pipeline Status

Stage	Status	Result
PDF extraction	Passed	18 pages extracted successfully
Text cleaning	Passed	Cleaned text saved successfully
Metadata update	Passed	P05 added to papers.csv
Chunking	Passed	chunks.jsonl generated after metadata update
DPR embeddings	Passed	Passage embeddings generated
FAISS index	Passed	56 vectors indexed
Retrieval	Passed	Top-5 chunks returned
RAG generation	Blocked	OpenAI API returned 429 credit_balance_exhausted

7. Retrieval Evaluation

Question: What are the main attention networks described in the paper?

- The system returned five chunks from the correct paper.
- The title, DOI and page metadata were present and correct for the test paper.
- The retrieved chunks were generally relevant to the question.
- Ranking was not perfect: Rank 2 was more directly relevant to the question than Rank 1, so ranking quality could potentially be improved.

8. Issues Found

Empty chunk dataset: The test paper was initially missing from papers.csv. chunks.jsonl therefore contained no chunks and build_index.py raised "No chunks found in chunks.jsonl."

Metadata dependency: The paper had to be registered in papers.csv before chunking could succeed.

Retrieval ranking: Relevant chunks were found, but the most directly useful chunk was not ranked first.

.env configuration: The .env file initially contained only the key value. The expected format is OPENAI_API_KEY=.

OpenAI credits: Generation reached the API successfully but returned 429 credit_balance_exhausted, so answer-quality evaluation is still blocked.

9. Remaining Work

- Rerun RAG answer generation after project API credits are available.
- Check whether the generated answer is supported by retrieved evidence.
- Check whether the answer is understandable for a non-expert user.
- Run additional questions, including at least one insufficient-evidence case.
- Test the team-provided dataset as required.
- Update the evaluation report and commit/push the final results to animish-evaluation.

10. Current Conclusion

The ingestion and retrieval stages of the WILL-RAG pipeline worked successfully with the added P05 research paper. The pipeline completed through PDF extraction, cleaning, chunking/metadata, DPR embeddings, FAISS indexing and retrieval. Final RAG answer generation is currently blocked by the OpenAI project credit balance rather than by a confirmed code failure.

Contribution percentage: To be confirmed by the project team.